Three questions, one reading journey

A ten-page reading guide with definitions in context. Expanded vocabulary is available in the companion HTML and PDF.

The question behind this research

What would let an AI do more than produce a convincing answer: understand a situation, anticipate consequences, choose an action, and remain under meaningful control? R06 approaches that question through three strands: **world models**, which represent and predict an environment; **neurosymbolic systems**, which combine learned patterns with explicit structure; and **independent safety assessment**, which examines evidence about capabilities and risks. These are connected questions, not one settled design.

Follow a single example

Imagine an assistant helping a robot put a cup on a shelf. It needs to recognize the cup, represent the shelf and obstacles, anticipate what different movements would do, and select an action. It must also recognize uncertainty and respect limits on what it may move. This is an illustrative teaching example, not a result reported by a source. Keep it in mind as the terminology becomes more technical.

What kind of evidence are you reading?

A **position paper** argues for a research direction. A **survey** organizes other work. An **experiment** tests a method under specified conditions, often using a **benchmark**, a repeatable task or test suite. A **replication** asks whether a result can be reproduced independently. The research compendium notes that R06 includes many proposals and blog posts. A compelling explanation therefore deserves a different level of confidence from repeated experimental evidence. [06.3, 06.5, 06.7, 06.11]

How to use this guide

Read the narrative first. Bold terms are explained where they appear; small vocabulary boxes reinforce distinctions used again later. The HTML companion and separate vocabulary PDF provide all 389 existing term-note entries grouped beside the corresponding learning topics. These are optional reference entries, not 389 claims newly verified for this guide. Listening prompts suggest where to pause when an idea arises in the podcast. For one primary reading, start with the DreamerV3 paper introduced on page 10. [06.5]

Capability	Evidence	Hypothesis
What a system can do under specified conditions.	Observations or results that support a particular claim.	A proposed explanation or mechanism that still needs testing.

LISTEN / PAUSE / LOOK

When the podcast makes a broad claim about future intelligence, ask: is this a proposed architecture, a demonstrated result, or an interpretation?

Inherited source references: [06.3, 06.5, 06.7, 06.11]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

PART 1 / Representing the situation

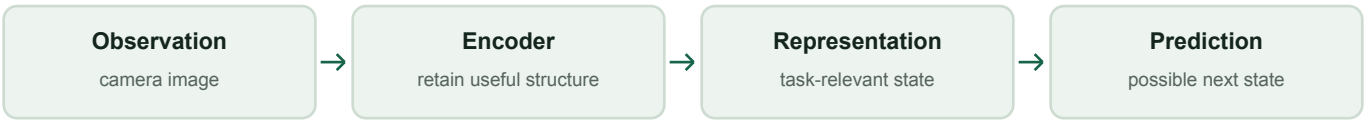
From observations to a world model

Keep what matters for the next decision

A **world model** is an internal model used to predict what might happen, including after a candidate action. A **state** describes the situation; a **representation** is the form in which the model encodes it. The robot does not necessarily need a perfect reconstruction of every pixel. It may need the cup position, shelf height, obstacles, and information about whether the cup is being held. Which details matter depends on the task. [06.3]

A representation is not automatically an understanding

A **latent state** is an internal, usually compressed representation rather than a directly observed measurement. An **encoder** converts observations into that representation. Compression alone does not guarantee that the model preserved the right physical relationships. A model may learn correlations that work in its training setting and fail when objects, lighting, or actions change. Ask what the representation enables, not whether the language used to describe it sounds human.



Illustrative teaching schema; not a reproduction of a paper figure.

Two ways to predict

A **generative model** can produce observations such as images or video. A **Joint Embedding Predictive Architecture (JEPA)** predicts a representation of a target rather than requiring a reconstruction of every detail. An **embedding** is a numerical representation in a learned space. In LeCun's proposed approach, prediction in representation space is meant to retain useful structure while leaving irrelevant detail aside. This is an architectural argument, not proof that every representation-space model captures causality. [06.3]

Make room for uncertainty

Aleatoric uncertainty concerns variability in the environment; **epistemic uncertainty** concerns gaps in the model's knowledge. A moving object may genuinely have several possible futures, or the model may simply lack information. Those situations call for different responses. A useful model should not disguise either one with a single confident-looking prediction.

Observation	Abstraction	Latent variable
A signal available to the system, such as a camera image.	A representation that keeps selected structure and leaves out detail.	An internal variable that can represent unobserved factors or alternative outcomes.

LISTEN / PAUSE / LOOK

Pause at descriptions of a model that “understands the world.” What exactly is represented, what is predicted, and how would the claim be tested?

Inherited source references: [06.3, 06.2]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

Learning useful representations

Where the learning signal comes from

Self-supervised learning (SSL) obtains a learning task from the data itself, for example predicting one part from another, rather than requiring a person to label every example. A model can observe how a scene changes and learn relationships between views or times. That is different from **reinforcement learning**, where behavior is improved using a reward or cost signal associated with actions. Observation and action can supply complementary information. [06.3]

The easy answer can be useless

An **objective** specifies what the learning procedure tries to improve; a **loss** expresses an error to reduce. If two encoders are rewarded only for producing matching representations, they could emit the same constant for everything. Their outputs would agree while distinguishing nothing. This is **representation collapse**. Good training must make predictions useful without allowing that shortcut.

Contrast and regularization

Contrastive learning distinguishes related examples from selected alternatives. **Non-contrastive learning** can instead use additional constraints to preserve informative representations. A **regularizer** discourages an undesirable or degenerate solution. VICReg, discussed in the existing notes, combines agreement with constraints on the spread of representation components and their correlations. **Variance** describes spread; **covariance** describes how components vary together. These are mechanisms for avoiding particular failures, not a universal guarantee of useful reasoning. [06.3]

Energy is a score here, not electricity

An **energy-based model (EBM)** assigns a compatibility score: compatible configurations receive low energy and incompatible ones higher energy. This use of “energy” is mathematical; it does not measure watts or carbon emissions. A **gradient** describes how a quantity changes locally with its inputs or parameters. **Differentiability** makes gradient-based optimization possible, but does not ensure that the objective represents what the user actually wants.

The founder's question

A learning curve can show that a numerical objective improved while leaving the practical question unanswered. Ask which information the representation must preserve, how collapse or shortcuts were ruled out, and whether the improvement transfers to the intended task. Separate the training objective from the outcome you care about.

Parameters
The adjustable values learned during training.

Inductive bias
A design preference that makes some solutions easier to learn than others.

VICReg
Variance-invariance-covariance regularization: a representation-learning criterion discussed in the notes.

LISTEN / PAUSE / LOOK
When training is described as “learning without supervision,” distinguish learning without manual labels from learning without any objective or design choices.

Inherited source references: [06.3]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

PART 1 / Predict, compare, act, then reconsider

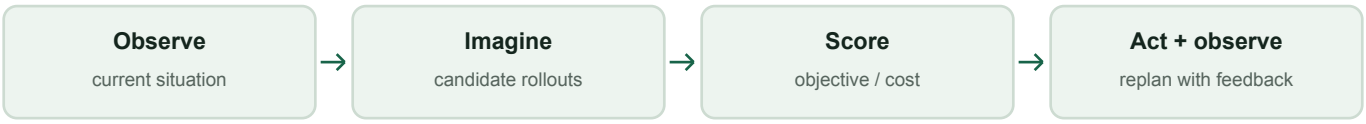
How a model becomes a plan

A prediction becomes useful when it changes a decision

A **policy** maps the available state or observation to an action. A **planner** considers alternatives before choosing one. In **model-predictive control (MPC)**, the system predicts the consequences of candidate action sequences, scores them, executes the first action of a preferred sequence, and plans again using the updated situation. The **planning horizon** is how far ahead it considers. Replanning is also called **receding-horizon control**. [06.3]

The actor, the critic, and the objective

An **actor** proposes or selects actions. A **critic** estimates their value or future cost; it is a learned evaluator, not an independent human auditor. An **action sequence** is a candidate series of moves. A **rollout** is a sequence of predicted states under such moves. The planner can compare rollouts, but it remains limited by the model and by the scoring rule. Optimizing a poor objective can produce a very effective wrong answer.



Illustrative teaching schema; not a reproduction of a paper figure.

Why longer plans can be difficult

Prediction mistakes can accumulate. Separately, when optimization differentiates through many repeated model steps, gradients can become extremely large or small: **exploding or vanishing gradients**. A **local minimum** is a nearby solution that looks better than its neighbors but need not be globally best. These are distinct difficulties. The existing notes caution that the podcast's "compounding error" explanation does not capture the specific gradient and optimization problems discussed in the GRASP post. [06.9]

Read GRASP as a specific proposal

The existing GRASP notes describe optimizing intermediate states alongside actions, adding noise to explore alternatives, and modifying gradient flow. Optimizing those intermediate states is called **collocation** or lifted-state planning. GRASP does not simply abandon states. Ask which failure it addresses, which experimental setting supports the result, and which assumptions remain. [06.9]

Cost function
A rule assigning a penalty to an outcome or trajectory.

Hierarchical planning
Planning through levels of goals and subgoals.

Feedback
New observations used to revise the next action.

LISTEN / PAUSE / LOOK
Pause when a planner is called more intelligent or reliable. Identify its objective, horizon, feedback, and the conditions under which it was tested.

Inherited source references: [06.3, 06.9]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

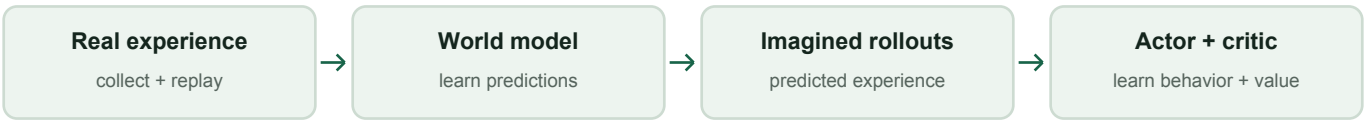
Dreamer: learning through imagined experience

Keep the three components separate

DreamerV3 is described in the existing material as a **model-based reinforcement-learning** system: it learns a world model and uses predicted experience to improve behavior. The **world model** predicts outcomes; the **actor** chooses actions; the **critic** estimates value. “Imagination” here means computation over model states, not consciousness or human-like subjective experience. [06.5]

Real experience still anchors the system

A **trajectory** is a sequence of observations, actions, and outcomes. An **experience replay buffer** stores past trajectories so they can be reused during learning. The world model is trained from collected experience. The actor and critic can then learn from imagined rollouts inside it. **Sample efficiency** concerns how much real interaction is needed; it is different from computational efficiency, which concerns the processing needed to learn or act.



Illustrative teaching schema; not a reproduction of a paper figure.

Why the reward matters

A **reward** scores outcomes for the learning task. A **return** combines rewards over a sequence, often giving less weight to distant rewards through **discounting**. The model, reward definition, and evaluation setting together shape the behavior that emerges. An impressive score can be meaningful without establishing that the same system will pursue the right goals in an unfamiliar deployment.

What the compiled result says

The research compendium reports performance across more than 150 tasks with one configuration. A **hyperparameter** is a setting chosen for the learning procedure rather than learned as a model parameter. Using one configuration across varied tasks is a robustness claim worth understanding. It is not proof of universal competence, safety, or transfer to arbitrary robots. The guide has not rechecked the original experiments. [06.5]

Read numbers with their denominators

Ask what counted as a task, a success, an episode, and a comparison baseline. An **episode** is one bounded interaction run. A **baseline** is the comparison method. Some podcast success-rate interpretations require checking against the original evaluation; they are not repeated here as established results.

Model-free learning
Learning behavior without an explicit learned dynamics model used for planning or imagined training.

Generalization
Performance beyond the examples or conditions used for learning.

Evaluation protocol
The exact rules used to obtain and compare results.

LISTEN / PAUSE / LOOK
When you hear the Dreamer example, separate learning in a model, acting in an environment, and measuring success. Those are three different stages.

Inherited source references: [06.5]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

Embodied intelligence: bodies, users, and transfer

A body makes assumptions consequential

Embodied AI perceives and acts through a virtual or physical form. A robot, wearable assistant, and virtual avatar can have different senses, action possibilities, and failure consequences. A **sensor** measures something; an **actuator** produces an action. In the cup example, seeing a successful movement on screen and safely executing it with a real gripper are different achievements. [06.2, 06.4]

Physical and mental models

A **physical world model** represents objects, dynamics, and the consequences of actions. A **mental world model**, as discussed in the existing notes, represents aspects of a user's goals, beliefs, or intentions. **Theory of mind** is reasoning about another agent's mental state. A model's prediction about what someone wants remains an inference; it is not permission to act. **Mixed initiative** describes interaction where either the person or system may initiate a step.

Planning across levels

“Put the cup away” is a high-level goal. Reaching the shelf, positioning a gripper, and applying motor control operate at different levels. **Hierarchical control** connects those levels. A **vision-language-action model (VLA)** links visual and language inputs to actions. Such a policy may be combined with a world model and an outcome-scoring model. That is a design proposal to assess, not evidence that every combined system is dependable. [06.2]

The transfer problem

Simulation-to-real transfer, often called **sim2real**, concerns moving behavior learned in simulation to physical hardware. Real surfaces, friction, lighting, and sensor errors can differ from a simulator. This is one form of **distribution shift**: the conditions of use differ from those represented in training or evaluation. **Imitation learning** learns from demonstrations, including **teleoperation**, where a human remotely controls a robot. Demonstrations also have a limited coverage of situations.

A useful reading distinction

The embodied-agent material proposes several routes to better perception, planning, memory, and interaction. Keep the architecture diagram separate from the evidence for each component and from end-to-end deployment results. When you dip into a paper, look first for the actual task and environment, then ask which part of the proposed system was evaluated.

Zero-shot
Attempting a task without task-specific training examples in the claimed evaluation setting.

Partial observability
The system cannot directly observe every relevant aspect of the situation.

Working memory
Temporary state used during ongoing processing; different from lasting knowledge learned in weights.

LISTEN / PAUSE / LOOK
For a wearable or robot example, ask what the system sensed, what it was allowed to do, and whether it was tested on real hardware or only in simulation.

Inherited source references: [06.2, 06.4, 06.10]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

PART 2 / Combining learned patterns with explicit knowledge

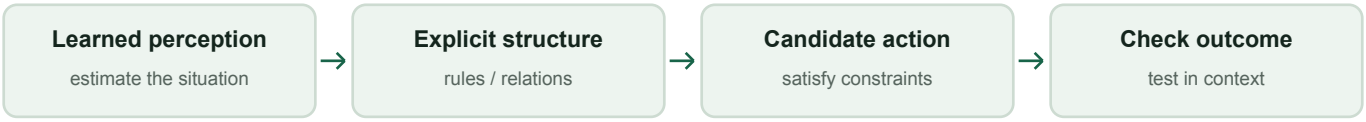
The alternative bet: smaller, structured intelligence

What “neurosymbolic” adds

Neurosymbolic AI combines neural learning with explicit structures such as rules, relations, or mathematical constraints. A **neural network** learns numerical parameters from data. A **symbolic representation** makes selected entities and relationships explicit. The existing notes present these approaches as potentially complementary to scaling, rather than proving that all large models should be replaced. [06.7]

A rule can reduce what must be learned

In the cup example, a rule that a blocked shelf cannot be used is explicit knowledge. A learned perception model could estimate whether the shelf is blocked; a separate constraint could govern the action. A **knowledge graph** represents entities and relationships as nodes and edges. A **logical constraint** states a relationship that an acceptable solution should satisfy. These structures can make particular requirements explicit, but incorrect rules or missing facts can still produce bad decisions.



Illustrative teaching schema; not a reproduction of a paper figure.

Structure is an inductive bias

An **inductive bias** favors some solutions over others. **Physics-informed neural networks (PINNs)**, for example, incorporate mathematical statements about physical behavior into learning. **Compositionality** is building more complex structures from reusable parts. **Few-shot learning** concerns learning from a small number of examples. The research question is whether the chosen structure reduces data or computation while preserving the capabilities the task needs. [06.7]

Distilling and checking

Distillation transfers aspects of a model's behavior or knowledge into another representation or model. **Symbolic distillation** seeks an explicit structure rather than only another network. **Formal verification** checks stated properties against a formal specification under defined assumptions. Passing such a check does not establish every real-world safety property, especially if the specification or connection to the physical world is incomplete.

Judge a bounded claim

Scaling laws describe empirical relationships between resources and measured performance. They are not guarantees about every useful capability. Equally, “smaller” is not a synonym for safer, more accurate, or cheaper in every setting. Ask: smaller by which measure, better on which task, compared with which baseline, and at what total system cost? Keep specific numerical comparisons tied to their original evaluation conditions.

Parameters versus rules
Learned numerical values and explicitly stated relationships play different roles.

Specification
A precise statement of the properties a system is intended to satisfy.

Interpretability
How a person can understand selected aspects of a system; it is not itself proof of correctness.

LISTEN / PAUSE / LOOK
When the podcast presents an alternative to scaling, identify what structure is added and what evidence would show it helps beyond one selected demonstration.

Inherited source references: [06.7]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

Reading independent safety evidence

What an assessment can tell you

An **evaluation** tests a capability or behavior under specified conditions. An **independent assessment** is conducted with some separation from the developer, but its value still depends on access, methods, incentives, and scope. A **research synthesis** brings together findings from multiple sources; it is not one new experiment. The research compendium uses the International AI Safety Report 2026 as a synthesis of capability developments and emerging risks. [06.11]

Capability does not come only from model size

Pre-training learns broad patterns from a large training dataset and produces a base model. **Post-training** further adapts it, for example through supervised fine-tuning, reinforcement learning, or distillation. **Inference** is running a trained model. **Inference-time compute** is processing used while producing an answer or taking action. The compiled safety summary highlights post-training as an increasingly important source of capability gains. [06.11]

The benchmark-to-deployment gap

A benchmark score answers the question set by a particular test. A real deployment may involve different tools, longer tasks, unreliable inputs, and consequences not represented by that test. **Data contamination** occurs when evaluation material or related answers enter training data. **Evaluation awareness** concerns a system recognizing test conditions; **gaming** exploits features of an evaluation without achieving its intended purpose. These are reasons to examine how a score was obtained, not reasons to dismiss all benchmarks.

Break the conclusion into parts

Separate **capability** (what can be done), **reliability** (how consistently it is done), and **safety** (whether relevant harms are acceptably controlled in context). A system might be capable but unreliable, or reliably pursue an inappropriate objective. A **threat model** states the actors, assets, and failure scenarios being considered. A test result without that context is easy to overgeneralize.

Questions worth taking into the paper

Look for the task, model version, access and tools, comparison baseline, success definition, and conditions excluded from the claim. Identify the gap between the reported result and the use you are considering. This guide is teaching how to read existing conclusions; it has not rerun experiments or independently checked the original report.

Failure mode A particular way a system can fail.	**External validity** How far a result applies beyond the setting in which it was obtained.	**Uncertainty** A limit on what the available evidence allows you to conclude.

LISTEN / PAUSE / LOOK

When a striking percentage appears, pause and identify its denominator and test conditions. Do not carry it directly into a founder presentation without checking its source.

Inherited source references: [06.11]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

Agents, control, and deployment boundaries

An agent adds an operating loop

An **AI agent** combines a model with a loop that can plan steps, use tools, observe results, and continue toward a goal. **Autonomy** concerns which decisions and actions it can take without human intervention. More tool access can expand both capability and the consequences of errors. A text-only answer, a code-executing assistant, and a robot moving an object present different problems. [06.11]

Useful categories of risk

The existing safety notes distinguish **malicious use**, where someone deliberately uses a system to cause harm; **malfunctions**, where the system fails or behaves unexpectedly; and **systemic risks**, which arise through widespread interaction with society and the economy. The categories can overlap. A strong capability can assist a defender or an attacker; the surrounding access and incentives matter.

Boundaries need implementation

Access control determines which resources a system may use. **Least privilege** gives it only the access needed for the task. A **sandbox** limits where code can run and what it can affect. A **human approval gate** reserves a specified action for a person. These are standard explanatory concepts, not claims that the R06 systems implement them successfully. A model saying it will obey a limit is different from an external mechanism enforcing that limit.

Memory is not automatic learning or accountability

A **context window** holds material available for the current model computation. A **key-value cache (KV-cache)** reuses internal computations over context; it is not a lasting record of verified knowledge. **Retrieval** brings stored material into a model's input. **Persistent memory** stores information across interactions. Any such memory can contain stale, incorrect, or untrusted material. **Provenance** records where information came from and how it was transformed.

Keep proposals labelled

The existing notes contain vendor and podcast ideas about composite architectures, agent identity, and hard constraints. They can prompt useful questions, but they should remain labelled as proposals. The compendium explicitly excludes the vendor memorandum from its core summary. Do not treat an appealing design story as an independently demonstrated safety guarantee. [06.1, 06.3]

Alignment
The problem of making a system's behavior fit intended goals and constraints.

Incident reporting
Recording observed failures so they can be investigated and learned from.

Monitoring
Observing behavior during operation; its usefulness depends on what it can detect and what follows.

LISTEN / PAUSE / LOOK
When an agent is described as safe, ask what is enforced outside the model, what is merely requested in a prompt, and what evidence covers the whole operating loop.

Inherited source references: [06.11, 06.3, 06.1]. Expanded vocabulary: see the same topic in the HTML companion or separate vocabulary PDF.

A reading route and one paper to start with

Bring the three questions back together

World models ask how a system can represent and anticipate its environment. Structured approaches ask what explicit knowledge can contribute alongside learning. Safety assessment asks what observations justify confidence in a particular use. None of these questions is settled by a polished demonstration alone. For a founder, the skill is moving between the mechanism, the evidence, and the decision without turning an attractive hypothesis into a promise.

One recommended first paper: DreamerV3

Start with **Mastering Diverse Domains through World Models** by Danijar Hafner and colleagues. It provides a concrete example of learning a world model and using imagined experience to improve behavior. Read the abstract and introduction, then inspect the method overview and results. Ask: what comes from real experience, what happens inside the model, and how is improvement measured? It is an anchor for the technical strand, not a substitute for the safety assessment or the neurosymbolic work. [06.5]

Use one stopping question at a time

After reading a topic, listen for its corresponding idea in the podcast. Pause when a claim becomes important or confusing, follow the relevant source bookmark, and look for the passage, figure, or evaluation that addresses it. You do not need to read every paper cover to cover before resuming. Explain the idea in your own words, then note what you still cannot establish. These are suggested reading actions; this guide has not performed them for you.

Keep the existing cautions visible

Podcast narratives can simplify mechanisms, omit evaluation conditions, or repeat a disputed numerical interpretation. Check important percentages, study attributions, and deployment claims against the cited source before relying on them. This guide draws on an existing compendium and vocabulary collection; it does not independently verify every underlying result. The expanded reference entries retain their inherited wording and qualifications.

Ask about mechanism	Ask about evidence	Ask about application
What would make the claimed effect happen?	Which result supports this exact claim, under which conditions?	What changes when the setting, user, tools, or consequences change?

LISTEN / PAUSE / LOOK

Before moving on from R06, explain the robot example using a world model, a policy, a critic, a constraint, and an evaluation. Then identify one thing you would still want to check.

Source bookmarks - links inherited from the existing bibliography

06.2 Embodied AI Agents: Modeling the World
06.3 A Path Towards Autonomous Machine Intelligence
06.4 World Model for Robot Learning
06.5 Mastering Diverse Domains through World Models
06.7 Neurosymbolic AI as an antithesis to scaling laws
06.9 Gradient-based Planning for World Models
06.10 Whole-Body Conditioned Egocentric Video Prediction
06.11 International AI Safety Report 2026

Scope: this guide compiles existing summaries and vocabulary notes; it is not a new systematic review. Source 06.1 is a vendor memorandum; the compendium records unavailable text for 06.6 and 06.8. Listening prompts are conceptual, not timestamped.